AI.208

AI for Robotics

COURSE OVERVIEW

Integrating AI methods—perception, planning, learning—into embodied robotic systems.



ILLUSTRATIVE HERO IMAGERY

AI FOR ROBOTICS

1

SESSION

60 MINUTES

ORIENTATION: EMBODIED AI

KEY TOPICS

- AI in the robot loop
- Embodiment and situatedness
- Perception–action coupling
- Course roadmap

TEACHING FOCUS

Put intelligence inside a body.

▲ **KEY TAKEAWAY** Embodied AI acts in the world it perceives.

2

SESSION

60 MINUTES

PERCEPTION FOR ACTION

KEY TOPICS

- Vision for grasping
- Tactile and force sensing
- Multimodal fusion
- Perception–action loop

TEACHING FOCUS

Let perception guide immediate action.

KEY TAKEAWAY Seeing is for doing.

3

SESSION

60 MINUTES

LEARNING FOR CONTROL

KEY TOPICS

- Imitation learning
- Reinforcement learning for control
- Sim-to-real
- Skill acquisition

TEACHING FOCUS

Teach robots skills rather than code them.

▲ **KEY TAKEAWAY** Robots learn best by doing.

4

SESSION

60 MINUTES

TASK & MOTION PLANNING

KEY TOPICS

- Task planning
- Motion planning integration
- Symbolic and geometric reasoning
- Manipulation planning

TEACHING FOCUS

Plan both what to do and how to move.

KEY TAKEAWAY Tasks give purpose; motion gives movement.

5

SESSION

60 MINUTES

HUMAN-IN-THE-LOOP AI

KEY TOPICS

- Teleoperation and shared autonomy
- Robot learning from humans
- Interactive learning
- Trust and transparency

TEACHING FOCUS

Blend human and machine intelligence.

KEY TAKEAWAY Humans and robots learn together.

6

SESSION

60 MINUTES

AI FOR ROBOTICS & REVIEW

KEY TOPICS

- Integrated robot demo
- Course recap and assessment
- Toward swarm robotics (AI.303)
- Capstone preview

TEACHING FOCUS

Run a full AI–robot pipeline.

KEY TAKEAWAY Embodied intelligence is the sum of perception, planni